

Total No. of Questions : 8]

SEAT No. :

P1020

[Total No. of Pages : 2

[4457]-213

S.E. (Computer Engineering) (Semester - I)
DIGITAL ELECTRONICS AND LOGIC DESIGN
(2012 Course)

Time : 2 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *Attempt Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6 and Q.No.7 or Q.No.8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*

Q1) a) Minimize the following function using K-map & realize using Logic gates.

$$F(A,B,C,D) = \sum m(1,3,7,11,15) + d(0,2,5) . \quad [4]$$

b) Convert following : [2]

$$(255)_{10} = (?)_{16}$$

c) Differentiate between standard TTL and CMOS logic circuit w.r.t. [6]

- i) Propagation delay
- ii) FANOUT
- iii) Figure of merit

OR

Q2) a) Convert the following numbers into binary and hexadecimal numbers. [4]

- i) $(46)_8$
- ii) $(20.5)_{10}$

b) Define the following terms and mention its standard values for TTL family. [6]

- i) Voltage and Current parameter.
- ii) Power Dissipation.
- iii) Noise margin.

c) Represent the following signed number in 2's complement method : [2]

- i) +8
- ii) -8

Q3) a) Design a 4-bit BCD to Excess-3 code converter circuit using minimum number of logic gates. [6]

b) Design Mod-5 synchronous counter using JK FFs. [4]

c) Draw the excitation table of S-R Flip-flop. [2]

P.T.O.

OR

- Q4)** a) Design a 3-bit binary to 3-bit gray code converter using IC-74138. [4]
b) How many Flip-flops are required to build a binary counter circuit to count from 0 to 2048. What is the frequency of the output of last Flip-flop for an input clock frequency of 6MHz? [6]
c) Perform the following : [2]
 $(1111)_2 + (1111)_2 = ?$
- Q5)** a) State and explain basic component of ASM chart? What is difference between ASM chart and conventional flow chart? [7]
b) Write VHDL code 8:1 Multiplexer using Behavioral and Dataflow modeling style. [6]

OR

- Q6)** a) Design a sequence generator circuit to generate the sequence 1-3-5-7 using Multiplexer Controller based ASM approach. [7]
Consideration :
i) If control input $C = 0$, the sequence generator circuit in the same state.
ii) If control input $C = 1$, the sequence generator circuit goes into next state.
b) Explain the following statements used in VHDL with suitable examples:[6]
i) Process.
ii) CASE.
iii) With-Select-When.
- Q7)** a) What are different types of PLDs? Design 3:8 decoder using PLD. [7]
b) Draw and explain the basic architecture of FPGA. [6]

OR

- Q8)** a) A combinational circuits is defined by the function [7]
 $F_1(A,B,C) = \sum m(0,1,2,4)$
 $F_2(A,B,C) = \sum m(1,3,5,6)$
Implement this circuit with PLA.
- b) A combinational circuits is defined by the function [6]
 $F_1(A,B,C) = \sum m(0,1,3,4)$
Implement this circuit with PAL.

